3 **(a)** **(i)** Define *force*.

.....

.....[1]

(ii) State *Newton's third law of motion*.

.....

.....

..... [2]

For Examiner's Use

(b) Fig. 3.1 shows the variation with time *t* of a jumping flea's acceleration *a*. The acceleration *a* is measured in unit of *g*, the acceleration of free fall. The flea of mass 210 μg jumped at nearly vertical take-off angle from ground.

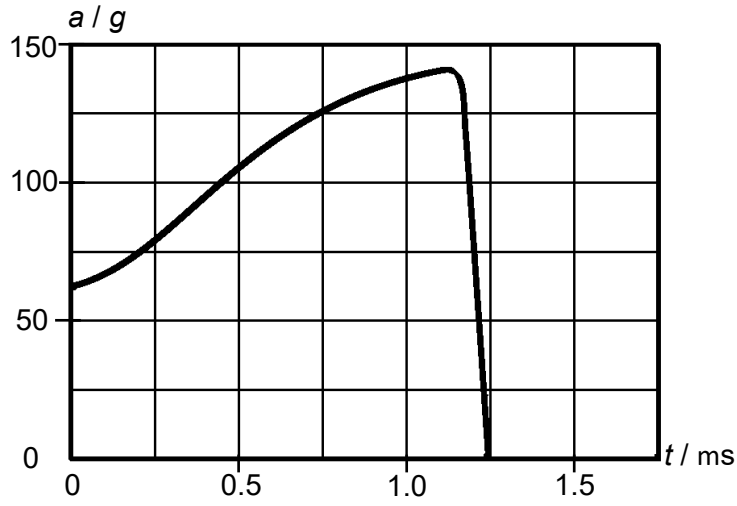

Fig. 3.1

(i) Use Fig. 3.1 to

1. determine the maximum net external force acting on the jumping flea,

force = ... N [2]

2. estimate the maximum speed achieved by the flea.

speed = ... m s^{-1} [3]

(ii) State and explain whether linear momentum is conserved during the take-off of the flea from the ground.

.....

.....

.....[2]